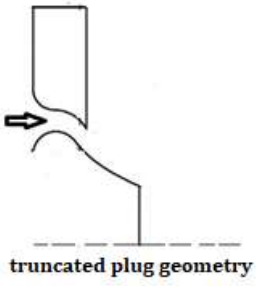
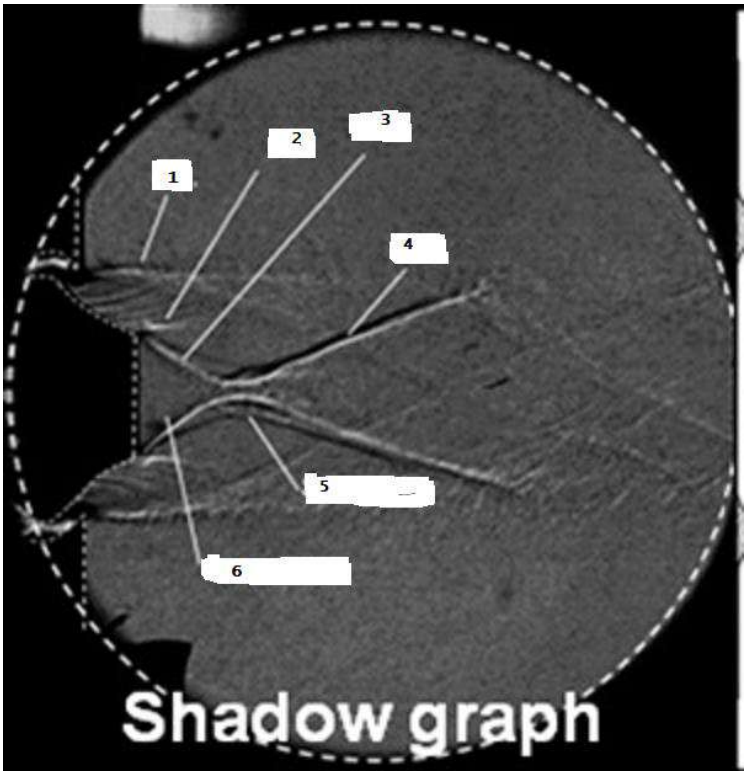
25.01.2023 - Gas Dynamics - MSAR - B. Favini

1. A supersonic steady two-dimensional uniform stream at mach number **M = 1.3** is flowing along a flat plate where impinges an oblique shock. Define the relative pressure-deflection polar and determine the different possible oblique shock reflections (assume ideal gas model). Describe the identified different reflections in the physical (x,y) plane. **(score 13/30)**

2. A piston located at the left boundary of a very long pipe moves to the right with parabolic law $\mathbf{x = t^2}$ up to **t = 1** and after this time with linear law **x = 2t - 1**. The gas inside the pipe is at rest with pressure **p = 1** and density $\boldsymbol{\rho = \gamma}$. Determine the evolution of the field and its dependence on the gas molecular weight (properties $\boldsymbol{\gamma}$ and **R)**. Describe the flowfield on the space-time and pressure-velocity planes. **(score 13/30)**

3. Identify waves and regions numbered in the visualization of the supersonic steady flow generated by a two-dimensional truncated plug nozzle. **(score 4/30)**



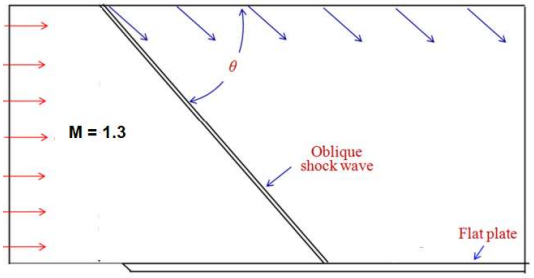


Gas Dynamics MSAR B. Favini written test 25.01.2023 solutions

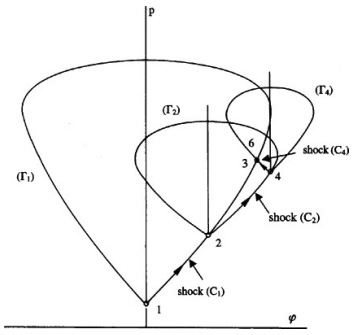
1. Reflection of an oblique shock impinging on a flat plate

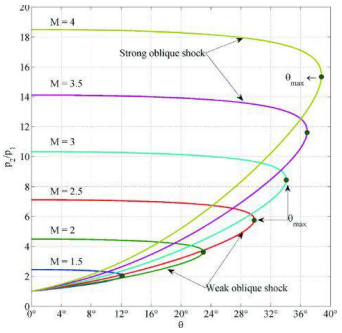
Question: "Define the relative pressure-deflection polar and determine the different possible oblique shock reflections"

The flow field

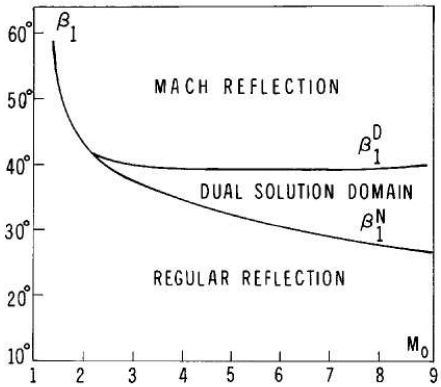


pressure-deviation polar (mail 17.11.2022)

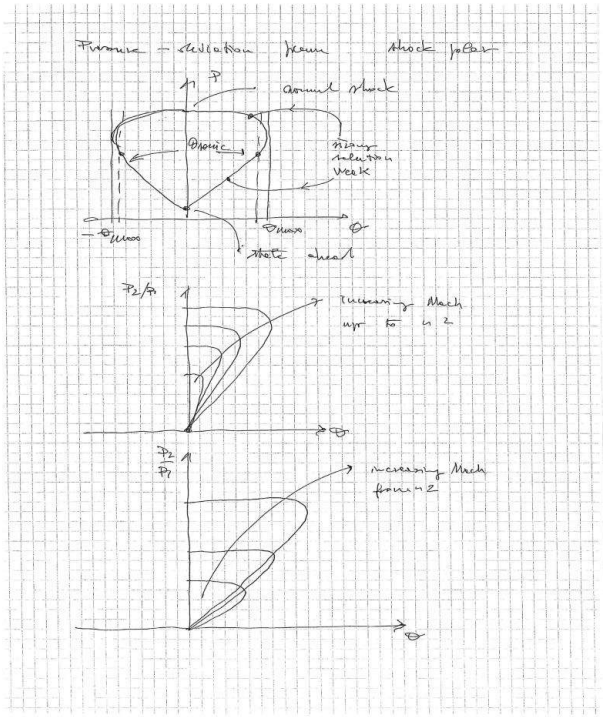




von Neumann and detachment angles – Mach plane (mail 21.11.2022)



Oblique shock polars (mail 17.01.2023)



From the analysis of the above reported images we can observe that when the Mach number of the incident oblique shock is small, its polar encloses the reflected oblique shock polar, the dual solution domain disappears and the Mach reflection case doesn't occur.

Indeed the computation of the state behind the incident oblique shock for different values of the deviation angle shows that the reflected oblique shock polar never intersects the incident shock polar and that for a deviation angle greater than roughly $\theta = 3.2$ the reflected oblique shock polars does not intersect the pressure axis.

When the deviation angle is greater than 3.2 there is no reflection and the impinging oblique shock curves smoothly and reaches the plate at 90° degree.